

ALFRED BRIGHT RIOCHI

AI/ML Engineer | Predictive Systems | Data Science

Nigeria (Remote / International) • github.com/Riochi511 • linkedin.com/in/riochi-ai453b9

PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with hands-on experience building and deploying end-to-end machine learning pipelines. Developed a Random Forest Classifier on 580,000+ real-world e-commerce records achieving a 0.9970 ROC-AUC score, deployed as a live Streamlit application on Hugging Face Spaces. Background in education and project management adds strong communication, analytical, and stakeholder management skills to technical capabilities. Seeking ML/AI internship roles where I can contribute to real-world predictive systems.

TECHNICAL SKILLS

Languages & Libraries: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

ML & Data: Random Forest, Classification, Time Series Forecasting, Prophet, NLP, TF-IDF, Cosine Similarity, RAG Pipelines, FAISS Vector Search, Sentence Transformers, LLM Inference, Feature Engineering, EDA, ROC-AUC, Model Evaluation

Deployment & Tools: FastAPI, Gradio, Streamlit, Hugging Face Spaces, Docker, GitHub Actions, Google Colab, Git, GitHub

PROJECTS

E-Commerce Conversion & Cart Abandonment Predictor

2025 – 2026

Capstone Project • TechCrush Academy • Deployed on Hugging Face Spaces

- Built an end-to-end ML pipeline on a 580,000-row Kaggle cosmetics dataset to predict purchase conversion vs. cart abandonment.
- Engineered 21 features from raw behavioural data; trained a Random Forest Classifier achieving 0.9970 ROC-AUC.
- Developed an interactive Streamlit web application enabling real-time prediction input and output visualisation.
- Containerised the application with Docker and deployed to Hugging Face Spaces; submitted to TekCrush GitHub organisation via pull request.
- Authored a full academic report (Chapters 1–5) covering problem definition, methodology, results, and business implications.

Beijing AQI Forecaster

2026

Time Series Forecasting • Facebook Prophet • [Live Demo](#) • Deployed on Hugging Face Spaces

- Built a PM2.5 air quality forecaster using Facebook Prophet trained on 41,757 hourly records from the Beijing PRSA dataset (2010–2014), capturing daily, weekly, and yearly seasonality.
- Evaluated on a 30-day holdout; MAE: 86.27 $\mu\text{g}/\text{m}^3$ — high variance reflects PM2.5 spike volatility, a known limitation of trend-based models.
- Deployed as a live Gradio app on Hugging Face Spaces with real-time hourly forecast generation; model artifact managed via Google Drive and gdown.

SmartJobMatch AI

2026

RAG Pipeline • Sentence Transformers • FAISS • TinyLlama • Streamlit • [Live Demo](#) • Deployed on Hugging Face Spaces

- Iterated from a TF-IDF + cosine similarity baseline to a full RAG pipeline using sentence-transformers (all-MiniLM-L6-v2), FAISS vector search, and TinyLlama-1.1B LLM — same 124,000-row LinkedIn dataset, fundamentally better architecture.

- Implemented PDF and DOCX resume upload with automatic text extraction (pypdf, python-docx); semantic search returns ranked matches with percentage scores.
- Resolved Git LFS deployment of large FAISS index and NumPy embeddings to Hugging Face Spaces free CPU tier; Space rebuilt live with branded Streamlit UI and AI career advice generation.

Cart Abandonment API & MLOps Pipeline

2025 – 2026

FastAPI · Docker · GitHub Actions CI/CD · Hugging Face Spaces

- Wrapped a trained cart abandonment prediction model in a FastAPI REST API and containerised it with Docker, deploying to Hugging Face Spaces.
- Built a GitHub Actions CI/CD pipeline that automatically builds the Docker image and tests the root endpoint on every push to main.
- Established a reproducible ML deployment pattern: train in Google Colab → export via joblib → serve with FastAPI → containerise with Docker → CI/CD via GitHub Actions.

EXPERIENCE

Secondary School Teacher

2022 – 2024

Britarch Secondary School · Nigeria

- Designed and delivered structured lesson plans for classes of 30–50 students, developing strong data-driven assessment and performance tracking habits.
- Managed academic records and student performance data, identifying patterns to personalise intervention strategies.
- Built communication and presentation skills through consistent delivery of technical content to non-technical audiences — directly transferable to stakeholder reporting in ML roles.

Building Project Coordinator

2021 – 2023

Independent · State-level Projects · Nigeria

- Oversaw end-to-end coordination of construction projects including scheduling, resource allocation, and stakeholder communication.
- Applied structured problem-solving and project management skills to deliver outcomes on time and within scope.

EDUCATION

AI/ML Engineering Programme

2025 – 2026

TechCrush Academy · Certificate of Completion · Awarded 2026

- Completed full ML pipeline development, model deployment, academic project reporting, and professional brand development.

B.Sc. Environmental Science & Resource Management

Graduated 2021

National Open University of Nigeria (NOUN)

- Developed analytical and quantitative research skills applied to large-scale resource and environmental datasets — directly transferable to ML data pipelines.

WASSCE

Secondary School Leaving Certificate

Britarch Secondary School · Nigeria

CERTIFICATIONS & PROFILES

TechCrush Academy · Certified AI/ML Engineer (2026) • [Live Project on Hugging Face Spaces](#) • [GitHub: Riochi511](#)